

Pinout

Firmware v1.4.0 | ESP32 DevKit V1 30-pin + Expansion Board HW-777 | 1 2569

1. ESP32 GPIO → Sensor / Component Connections

GPIO Pin	Component	Pin	
GPIO 4	DS18B20 Waterproof	DATA	External Pull-up 5.1kΩ VCC–DATA ESP32 (sensor)
GPIO 32	DHT22	DATA (Pin 2)	Pin 1=VCC, Pin 2=DATA, Pin 3=NC, Pin 4=GND — 3
GPIO 33	Buzzer Module (Active)	I/O	Pin →: GND – I/O – VCC
GPIO 34	Capacitive Soil Moisture	Signal (S)	ADC1 Input-Only — Output Pull-up
GPIO 21	LCD I2C 16x2	SDA	I2C Default ESP32 — Address 0x27 — 5V (3.3V)
GPIO 22	LCD I2C 16x2	SCL	I2C Default ESP32 — SDA (4: VCC, GND, SDA, SCL)
GPIO 26	Relay Module CH1	IN1	CH1 = — manual/schedule (auto mode)
GPIO 27	Relay Module CH2	IN2	CH2 = — Dashboard OFF
GPIO 14	Relay Module CH3	IN3	CH3 = 220V AC — auto: ≥ 35°C
GPIO 25	Relay Module CH4	IN4	CH4 = 24V DC — auto: Humidity < 60% (max 5/)

GPIO12 Relay Strapping Pin (boot fail) | GPIO34 ADC1 Input-Only

2. Relay Module 4CH — Output Wiring

Relay Channel	Load	Terminal	/	Mode
CH1 GPIO 26	()	NO → COM	Dashboard (manual/schedule)	manual
CH2 GPIO 27	—	—	Dashboard — OFF	—
CH3 GPIO 14	220V AC ×2	NO → COM → L	AC — Neutral Relay	auto
CH4 GPIO 25	24V DC	NO → COM	Boost Converter OUT– – (Common GND — Relay)	auto

⚠ Active-LOW: LOW=ON, HIGH=OFF | NO (Normally Open), NC

3. Power Distribution

From	To	Terminal /	Wire Gauge	Notes
220V AC Line	S-120-12 PSU (12V/10A)	L, N, PE	1.5 mm²	L=Phase, N=Neutral, PE=Earth
S-120-12 (+V=12V, -V)	XL4015 Step-Down (IN+, IN-)	IN+, IN-	1.0 mm²	Input 12V → Output 5V Trimpot 5.0V ESP32
XL4015 Step-Down (OUT+=5V)	ESP32 Expansion Board HW-777	OUT+ → 5V, OUT- → GND	0.5–1.0 mm²	ESP32, Relay IN, Buzzer, Fan 5V, LCD VCC (logic)
S-120-12 (+V=12V, -V)	Boost Converter (IN+, IN-)	IN+, IN-	1.5 mm²	Input 12V → Output 24V
Boost Converter (OUT+, OUT-)	Relay CH4 (COM)	OUT+ → COM	1.5 mm²	Trimpot Boost 24V Relay
Relay CH4 (NO)	DC 24V (+)		1.5 mm²	24V Relay → +
Boost Converter (OUT-)	DC 24V (-)		1.5 mm²	GND Relay — Common GND

⚠ Common GND — component GND | Boost Converter

4. Pinout

4a. DHT22 — + (GPIO 32)

Pin	Signal	Connection
Pin 1	VCC	3.3V or 5V (Expansion Board 5V rail)
Pin 2	DATA	GPIO 32
Pin 3	NC	(No Connect) —
Pin 4	GND	GND (Common Ground)

4b. DS18B20 Waterproof — (GPIO 4) — Flat Side

Pin Header (Flat Side)	Pin Label	Pin Description
Pin 1 (Left)	GND	GND (Common Ground)
Pin 2 (Middle)	DATA	GPIO 4 + Pull-up 5.1kΩ to VCC
Pin 3 (Right)	VCC	3.3V or 5V

⚠ -127°C | Pull-up 5.1kΩ to VCC | 5m PVC Conduit